

Exercise sheet for Week 11

♪ **Exercise 1.** For the complex numbers $z = 1 - i$ and $w = 2 + 3i$, express the following quantities in the form $x + iy$ for $x, y \in \mathbb{R}$.

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|------------------|------------------|--------------|-------------|
| (a) $z + w$, | (b) $z - w$, | (c) zw , | (d) w^2 , |
| (e) $z\bar{w}$, | (f) $\bar{w}z$, | (g) $13/w$, | (h) z/w . |

♪ **Exercise 2.** Show that for any complex number z ,

- (i) $\Re(z) = \frac{1}{2}(z + \bar{z})$,
- (ii) $\Im(z) = \frac{1}{2i}(z - \bar{z})$,
- (iii) $\Re(iz) = -\Im(z)$.

♪ **Exercise 3.** Find $|z|$, $\arg(z)$, $\ln(z)$ for each of the following complex numbers z .

- (a) $-1 - i$, (b) $1 + \sqrt{3}$, (c) $-6i$, (d) $(-2 + 2i)^3$, (e) $-3 - \sqrt{3}i$.

♪ **Exercise 4.** In each of the following cases, give an example of a complex number $z = x + iy$ with the property stated. Here $\tan^{-1} : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$ is the inverse of the function $\tan : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$.

- (i) $\arg(z) = \tan^{-1}(y/x) + \pi$,
- (ii) $\arg(z) = \tan^{-1}(y/x) - \pi$,
- (iii) $\arg(-z) = -\arg(z)$,
- (iv) $\arg(z^2) = \pi - \arg(z)$.

♪ **Exercise 5.** Find the square roots of (1) $1 + i$ and (2) $1 + i\sqrt{3}$.

♪ **Exercise 6.** Find all the roots of the complex polynomial $P(z) = z^6 - 1$. Write your answers both in polar form *and* in the form $z = a + ib$ for $a, b \in \mathbb{R}$.

♪ **Exercise 7.** Solve the following problems.

- (i) Express $\frac{1+i}{\sqrt{3}+i}$ in the form $x + iy$, where $x, y \in \mathbb{R}$.
- (ii) Write each of $1 + i$ and $\sqrt{3} + i$ in polar form.
- (iii) Using your results, deduce that

$$\cos \frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}}, \quad \text{and} \quad \sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

♪ **Exercise 8.** Solve the following problems.

- (i) Find the real and imaginary parts of $(\sqrt{3} + i)^{10}$ and $(\sqrt{3} + i)^{-7}$.
- (ii) For which integer values of n is $(\sqrt{3} + i)^n$ real?

♪ **Exercise 9.** Find a formula for $\sin 3\theta$ in terms of $\sin \theta$. Hence write down a cubic equation which has $\sin \frac{\pi}{18}$ as a root. What are the other roots of your equation?